

EV-PK Simulator: Synthetic Benchmark Datasets for Model Selection and Demotion

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Purpose

This document describes ten small, self-contained CSV datasets we use to exercise the end-to-end pipeline: five core benchmarks covering single-compartment (1c), two-compartment (2c), and three-compartment (3c) PK, plus edge cases that should trigger demotion (AICc and identifiability checks); and five add-on stress tests that probe guardrails (tail-AUC demotion, small- n AICc behavior, noise robustness, preprocessing hygiene, and per-row dosing). These datasets ship with our repo under `benchmarks/` (`core/` and `addons/`) and are intended for judges to reproduce our selection logic quickly.

General data-generation recipe

All datasets are IV bolus at $t=0$ with Dose = 100.000 mg (except `pk_3c_weak_tail_demote`, which uses 60.000 mg), species labels for context, and uniform time grids on $[0, 24]$ h. Concentrations are generated from closed-form macro-exponential solutions and then lightly noised to mimic measurement error:

$$\begin{aligned} \mathbf{1c:} \quad C(t) &= \frac{\text{Dose}}{V_d} e^{-k_{\text{el}} t} \\ \mathbf{2c:} \quad C(t) &= A e^{-\alpha t} + B e^{-\beta t}, \quad \alpha \neq \beta \\ \mathbf{3c:} \quad C(t) &= A e^{-\alpha t} + B e^{-\beta t} + C e^{-\gamma t}, \quad \alpha > \beta > \gamma \end{aligned}$$

Noise model: multiplicative log-normal with $\sigma \approx 5\%$ (then clipped at a small positive floor to avoid $\log(0)$ in log-scale fits). This keeps shapes diagnostic without obscuring ground truth.

Why these cases?

- **1c clear:** sanity check that the fitter prefers 1c and does not overfit.
- **2c clear:** classic biphasic decay with well-separated $\alpha \gg \beta$.
- **2c near-demote:** either $\alpha \approx \beta$ (rates nearly merged) or a vanishing B tail, so demotion to 1c is favored.
- **3c triphasic:** three well-separated phases; 3c clearly superior.
- **3c merged→demote:** two rates purposefully near-equal; our identifiability guard should demote 3→2.

Selection & demotion logic (as implemented)

When fitting, we compute goodness-of-fit metrics (AIC/AICc, R^2) and identifiability diagnostics. A 3c fit may be *demoted* to 2c if any of the following hold:

1. $\Delta\text{AICc} \equiv \text{AICc}_{3c} - \text{AICc}_{2c} > \Delta^*$ (default $\Delta^*=2$),
2. the minimum relative rate separation $\min\{|r_i - r_j|\} / \max(r_i, r_j)$ falls below a threshold (≤ 0.05 by default),
3. the slowest-phase tail AUC fraction is below a small threshold (default < 0.08).

Similarly, a 2c fit may be demoted to 1c when $\Delta\text{AICc} > 2$ or when $\alpha \approx \beta / B \approx 0$ makes the second phase non-identifiable. We also enforce a tiny minimum gap ($\text{MIN_GAP}=1\text{e-}3$) between ordered rates via a softplus parameterization to keep $\alpha > \beta > \gamma$ strictly ordered.

Scoring & diagnostic definitions (exact formulas)

We use *linear-space* residuals for the benchmarks (`AICc_linear`).

$$\begin{aligned}\text{AICc} &= 2k - 2 \ln \hat{L} + \frac{2k(k+1)}{n-k-1}, \quad \ln \hat{L} = -\frac{1}{2\sigma^2} \sum_i (y_i - \hat{y}_i)^2 + C \text{ (constant)} \\ \text{sep_rel} &= \min_{i < j} \frac{|r_i - r_j|}{\max(r_i, r_j)}, \quad r \in \{\alpha, \beta, \gamma\} \\ \phi_{\text{tail},3c} &= \frac{C/\gamma}{A/\alpha + B/\beta + C/\gamma}, \quad \phi_{\text{tail},2c} = \frac{B/\beta}{A/\alpha + B/\beta}\end{aligned}$$

Demotion thresholds (defaults): $\Delta\text{AICc} > 2$ (prefer simpler), $\text{sep_rel} \leq 0.05$ (rate merge), $\phi_{\text{tail},3c} < 0.08$ (weak tail).

Data schema & units

Unless stated otherwise, columns are:

- **species** (string), **subject** (int), **time** (real), **concentration** $\equiv$ `conc_mg_L`, and for the per-row dose add-on: `dose_mg_per_kg`, `weight_kg`.

Concentrations are in mg L^{-1} . Before log-space plots we clip by $C \leftarrow \max(C, \varepsilon)$ with $\varepsilon = 10^{-9} \cdot \max(C)$.

Time grid & randomness

Core sets use a uniform grid on $[0, 24]$ h with $n=42$ points and $\Delta t = 24/(42-1) \approx 0.585$ h, including both endpoints. Core noise $\sigma=0.05$; the high-noise add-on uses $\sigma=0.18$. Random seeds are fixed in the generator script and listed there for byte-for-byte regeneration.

Quick summary of the five CSVs

Table 1 summarizes simple diagnostics computed directly from each CSV (no model assumptions): n , time range, max observed C , a global log-linear half-life, and the ratio of early- vs late-phase log-slope magnitudes (bigger than one suggests multi-phase behaviour). The last column states the *expected* selected model under our criteria.

Case	File	Dose (mg)	n	$t_{\min}$ (h)	$t_{\max}$ (h)	$C_{\max}$	$t_{1/2}$ (global, h)	$ m_{\text{early}} / m_{\text{late}} $	Expect
1c clear (IV bolus)	pk_1c_ivbolus_clear.csv	100.0	42	0.00	24.00	2.538	3.48	0.95	1c (no demotion)
2c clear (classic biphasic)	pk_2c_biphasic_clear.csv	100.0	42	0.00	24.00	6.771	5.26	3.66	2c (no demotion)
2c near-demote ($\alpha \approx \beta$ / tiny B)	pk_2c_near_demote.csv	100.0	42	0.00	24.00	4.387	1.40	1.04	2c (likely demote)
3c triphasic (well-separated)	pk_3c_triphasic_clear.csv	100.0	42	0.00	24.00	7.531	6.13	6.30	3c (no demotion)
3c merged $\rightarrow$ demote (two rates $\sim$ equal)	pk_3c_merged_rates_demote.csv	100.0	42	0.00	24.00	5.656	4.48	1.80	Demote 3 $\rightarrow$ 2 ($\alpha \approx \beta$)

Table 1: Descriptive stats computed from the provided CSVs.

Per-dataset notes

1c clear (IV bolus)

Generated with a single exponential (V_d and k_{el} chosen to give $C(0) \approx \text{Dose}/V_d$ and $t_{1/2} \approx 3.48$ h). In our app the 1c fit yields the lowest AICc; 2c/3c do not beat it once AICc penalization kicks in.

2c clear (classic biphasic)

Parameters chosen such that $\alpha \gg \beta$ and both A, B are appreciable. Early/late slope ratio ≈ 2.8 confirms strong biphasic behavior. The 2c model should win decisively in AICc; 1c underfits (systematic early-time residuals), 3c adds an unnecessary tiny phase.

2c near-demote ($\alpha \approx \beta$ or tiny B)

Built to be barely distinguishable from 1c: either $\alpha \approx \beta$ (rates nearly merged) or $B \ll A$. Our selection typically demotes to 1c (or yields ΔAICc very near zero with a nudge toward parsimony).

3c triphasic (well-separated)

Three distinct phases with $\alpha \gg \beta \gg \gamma$. Early/late slope ratio ≈ 6.3 and a visible tri-phasic shoulder on semilog. 3c retains separation and wins AICc. Identifiability checks pass (min relative rate separation well above threshold).

3c merged $\rightarrow$ demote (two rates nearly equal)

Constructed with $\alpha \approx \beta$ so the min relative separation triggers the demoter and the fit collapses to an equivalent 2c tail. Expect a “Demoted (by rule)” badge in the UI with small ΔAICc favoring 2c (or at least failing to justify 3c).

Add-on stress tests (recommended)

These five additional CSVs exercise guardrails and edge cases not covered by the core set (tail-AUC demotion, small- n AICc behavior, noise robustness, preprocessing hygiene, and per-row dosing).

Case	File	n	Feature exercised	Expect
3c $\rightarrow$ 2c by weak tail	pk_3c_weak_tail_demote.csv	42	Tiny slow γ tail (low tail AUC)	Demote 3 $\rightarrow$ 2 ($\text{tail}_{auc_{min}}$)
2c biphasic — sparse	pk_2c_biphasic_sparse.csv	10	Small- n regime; AICc small- n correction	2c (no demotion)
1c clear — high noise	pk_1c_ivbolus_noisy.csv	42	Noise robustness ($\sigma \approx 0.18$)	1c (no demotion)
Data hygiene stress	pk_data_hygiene_issues.csv	—	Non-positive C , duplicate t , single outlier	2c fits; warnings only
Per-row dose (mg/kg)	pk_mgkg_per_row_dose.csv	2×21	hasDoseFromData + multi-subject	1c (each subject)

Table 2: Add-on CSVs that exercise guardrails and edge cases beyond the core five.

Add-on: 3c $\rightarrow$ 2c by weak tail (triggers tail_auc_min)

Analyze mode: select “three”; keep demotion on with default thresholds ($\text{tail_auc_min} = 0.08$).

What it proves: Even with acceptable rate separation, a negligible slow tail should demote 3 $\rightarrow$ 2 by the tail-AUC rule; 2c also wins AICc.

Badges to expect: small/negative ΔAICc favoring 2c; sep_rel OK; Demoted (by rule).

Add-on: 2c biphasic — sparse sampling

Analyze mode: select “two”.

What it proves: Clear biphasic signal can be recovered with only ~ 10 points; demonstrates AICc small- n behavior and wider CIs.

Badges to expect: 2c selected; reasonable n shown (10).

Add-on: 1c clear — high-noise robustness

Analyze mode: select “one”.

What it proves: Penalization prevents overfitting; 2c/3c do not justify extra parameters under $\sigma \approx 0.18$.

Badges to expect: 1c selected; residuals patternless.

Add-on: Data-hygiene stress test

Analyze mode: select “two”.

What it proves: Preprocessing drops non-positive concentrations, tolerates duplicate times, and remains stable with a moderate outlier.

Badges to expect: Warnings (non-positive filtered); 2c fit converges with finite CIs; no NaNs/Infs.

Add-on: Per-row dose (mg/kg) with weights (two subjects)

Analyze mode: select “one”.

What it proves: hasDoseFromData disables global dose and uses per-row dose/weight; each subject fits separately; Fit $\rightarrow$ Sim bridge remains correct.

Badges to expect: Two subjects listed; each yields a clean 1c fit with subject-specific parameters.

Preprocessing (data hygiene) policy

- Drop rows with non-positive concentration ($C \leq 0$) with a warning.
- If duplicate (subject, time) exist, aggregate by mean concentration before fitting (logged in warnings).
- Outliers are *kept*; robustness is handled by the fitting loss/UQ (no special-casing).

How judges can reproduce

1. From the home page, choose the **Mode** (one, two, or three compartments) appropriate for the dataset being tested.
2. Set **Parameter source** to *Estimate from data (fit)*.
3. Upload the CSV (drag-and-drop or *Choose file*).
4. Click **Fit**. The app will estimate parameters (per subject if the file contains multiple subjects).
5. *If your build shows them*, review badges/fields such as **AICc**, ΔAICc , n , **sep_rel**, and any warnings (e.g., filtered non-positive concentrations, model demotion). Otherwise, rely on the *Selected model* label and the plots.
6. Click **Download PDF report** to save a comprehensive report with key metrics and several plots.

Reproducibility notes

Rates are positive via softplus; for 3c we enforce an ordered triplet ($\alpha > \beta > \gamma$) with a minimum gap, and use a stable softmax split (s_1, s_2, s_3 with s_3 anchored) to avoid degenerate allocations among A, B, C . Noise seeds are fixed in the script for deterministic regeneration.

Files in this bundle

- Core CSVs (benchmarks/core/):
 - pk_1c_ivbolus_clear.csv
 - pk_2c_biphasic_clear.csv
 - pk_2c_near_demote.csv
 - pk_3c_triphasic_clear.csv
 - pk_3c_merged_rates_demote.csv
- Add-on CSVs (benchmarks/addons/):
 - pk_3c_weak_tail_demote.csv
 - pk_2c_biphasic_sparse.csv
 - pk_1c_ivbolus_noisy.csv
 - pk_data_hygiene_issues.csv
 - pk_mgkg_per_row_dose.csv
- Reports (one per CSV; same basenames):
Report(<basename>.csv).pdf (e.g., Report(pk_2c_near_demote.csv).pdf)
- Optional plots (PDF) illustrating log-linear behaviour for each case.